=== src/m3/prepare_03_sec72.html === <!-- UNIT 2: BARTLE §7.2 (T10-T19) --> <section class="page"> UNIT 2 · BARTLE §7.2 <h1>Riemann Integrable Functions</h1> <hr>
<!-- T10 (K10) --> T10<h2>The Cauchy Criterion for integrability</h2> Testing integrability without knowing the value L in advance (Bartle Theorem 7.2.1; PYQ MAT-21 Q8). Theorem 7.2.1 · Cauchy Criterion (Bartle p. 208) · say it word for wordA function f : [a,b] \to \mathbb{R} belongs to \mathcal{R}[a,b] if and only if for every \varepsilon > 0 there exists \eta_\varepsilon > 0 such that if \dot P and \dot Q are any tagged partitions of [a,b] with \lVert P\rVert < \eta_\varepsilon and \lVert Q\rVert < \eta_\varepsilon, then \lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon.
In plain words: f is integrable if and only if, whatever \varepsilon > 0 you are handed, you can find a mesh bound \eta_\varepsilon > 0 so that any two Riemann sums with mesh <\eta_\varepsilon are within \varepsilon of each other—without ever naming L. Hypotheses · Theorem 7.2.1 \lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon holds for all pairs of tagged partitions \dot P, \dot Q with \lVert P\rVert, \lVert Q\rVert < \eta_\varepsilonfor Dirichlet's function C7, two partitions with the same tiny mesh still give \lvert S(f;\dot P) - S(f;\dot Q)\rvert = \lvert 1 - 0\rvert = 1 \not< 1/2 completeness of \mathbb{R} (Cauchy Convergence Criterion 3.5.5 for sequences in \mathbb{R})used in the (\Leftarrow) direction to guarantee that the Cauchy sequence of numbers (S(f;\dot P_m)) converges to a real limit A \in \mathbb{R} Skeleton · use the Cauchy Criterion to prove f \in \mathcal{R}[a,b]Let \varepsilon > 0 be given. Choose \eta_\varepsilon = $? $ > 0. Let \dot P, \dot Q be any tagged partitions of [a,b] with \lVert P\rVert < \eta_\varepsilon and \lVert Q\rVert < \eta_\varepsilon. Then \lvert S(f;\dot P) - S(f;\dot Q)\rvert \le $? $ < \varepsilon. By the Cauchy Criterion (Theorem 7.2.1), f \in \mathcal{R}[a,b]. ∎ Know(\Rightarrow) f \in \mathcal{R}[a,b] with integral L. (\Leftarrow) For each \varepsilon > 0 there is \eta_\varepsilon > 0 making \lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon.Want(\Rightarrow) Produce \eta_\varepsilon. (\Leftarrow) Construct the limit A \in \mathbb{R} using \varepsilon = 1/n and prove \int_a^b f = A. Scratch · how (\Leftarrow) builds the missing number A Take \varepsilon = 1/n to get mesh bounds \delta_n > 0, forced decreasing (\delta_n \ge \delta_{n+1}). Pick one tagged partition \dot P_n with \lVert P_n\rVert < \delta_n. For m > n, both \dot P_m and \dot P_n have norm < \delta_n, so \lvert S(f;\dot P_m) - S(f;\dot P_n)\rvert < 1/n: the real sequence a_m = S(f;\dot P_m) is Cauchy! Call its limit A. Then any \dot Q with \lVert Q\rVert < \delta_K is within 1/K of S(f;\dot P_K), which is within 1/K of A; choose K > 2/\varepsilon. </section>
<section class="page"> Proof of Theorem 7.2.1 (Bartle p. 208) (\Rightarrow) If f \in \mathcal{R}[a,b] with integral L, let \eta_\varepsilon := \delta_{\varepsilon/2} > 0 be the number from Definition 7.1.1 corresponding to \varepsilon/2. If \dot P, \dot Q have \lVert P\rVert < \eta_\varepsilon and \lVert Q\rVert < \eta_\varepsilon, then \lvert S(f;\dot P) - L\rvert < \varepsilon/2 and \lvert S(f;\dot Q) - L\rvert < \varepsilon/2.Definition 7.1.1 applied twice with tolerance \varepsilon/2. By the Triangle Inequality,

\lvert S(f;\dot P) - S(f;\dot Q)\rvert = \lvert S(f;\dot P) - L + L - S(f;\dot Q)\rvert \le \lvert S(f;\dot P) - L\rvert + \lvert L - S(f;\dot Q)\rvert < \varepsilon/2 + \varepsilon/2 = \varepsilon.
Add and subtract L, then apply the Triangle Inequality (P05). (\Leftarrow) For each n \in \mathbb{N}, let \delta_n > 0 be the number \eta_{1/n} such that whenever \dot P and \dot Q have norms <\delta_n, we have \lvert S(f;\dot P) - S(f;\dot Q)\rvert < 1/n. We may assume \delta_n \ge \delta_{n+1} for all n (otherwise replace \delta_n by \delta'_n := \min\{\delta_1,\dots,\delta_n\}).Hypothesis applied with \varepsilon = 1/n; taking running minimums keeps \delta_n decreasing so finer partitions satisfy all earlier bounds. For each n \in \mathbb{N}, choose a tagged partition \dot P_n with \lVert P_n\rVert < \delta_n. If m > n, then \lVert P_m\rVert < \delta_m \le \delta_n and \lVert P_n\rVert < \delta_n, so
\lvert S(f;\dot P_m) - S(f;\dot P_n)\rvert < 1/n.
Both \dot P_m and \dot P_n have norm <\delta_n, so Step 1 applies to the pair (\dot P_m, \dot P_n). Consequently, the sequence (S(f;\dot P_m))_{m=1}^\infty is a Cauchy sequence in \mathbb{R}. By the Cauchy Convergence Criterion for sequences (Theorem 3.5.5, P06), this sequence converges in \mathbb{R}; let A := \lim_m S(f;\dot P_m).Completeness of \mathbb{R} (P06). Passing to the limit in Step 2 as m \to \infty gives \lvert S(f;\dot P_n) - A\rvert \le 1/n for all n \in \mathbb{N}.Order property of sequence limits (strict < 1/n becomes \le 1/n in the limit). To see that A is the Riemann integral of f, given \varepsilon > 0, choose K \in \mathbb{N} with K > 2/\varepsilon (so 1/K < \varepsilon/2). If \dot Q is any tagged partition with \lVert Q\rVert < \delta_K, then
\lvert S(f;\dot Q) - A\rvert \le \lvert S(f;\dot Q) - S(f;\dot P_K)\rvert + \lvert S(f;\dot P_K) - A\rvert < \frac{1}{K} + \frac{1}{K} = \frac{2}{K} < \varepsilon.
Since \varepsilon > 0 is arbitrary, f \in \mathcal{R}[a,b] with \int_a^b f = A. ∎Archimedean property (P13), Step 1 for (\dot Q, \dot P_K), Step 4 for n=K, and Triangle Inequality.   Worked example · Bartle Example 7.2.2(a) Let g(x) = 2 on [0,1] and 3 on (1,3] as in Example 7.1.4(b). For any \dot P, \dot Q with \lVert P\rVert < \delta and \lVert Q\rVert < \delta, we showed 8 - 5\delta \le S(g;\dot P) \le 8 + 5\delta and 8 - 5\delta \le S(g;\dot Q) \le 8 + 5\delta. Subtracting gives \lvert S(g;\dot P) - S(g;\dot Q)\rvert \le 10\delta. Given \varepsilon > 0, choose \eta_\varepsilon := \varepsilon/20; then \lVert P\rVert, \lVert Q\rVert < \eta_\varepsilon \Rightarrow \lvert S(g;\dot P) - S(g;\dot Q)\rvert \le 10(\varepsilon/20) = \varepsilon/2 < \varepsilon. By Theorem 7.2.1, g \in \mathcal{R}[0,3]. In the model paper: Q23 (PYQ MAT-21 Q8). </section>
<!-- T11 (K11) --> <section class="page"> T11<h2>Dirichlet's function (and relatives) are not integrable</h2> Negating the Cauchy Criterion to prove f \notin \mathcal{R}[a,b] (Bartle Example 7.2.2(b), Exercises 7.2.1–7.2.2; PYQ MAT-22 Q4, MAT-22 MCQ 17, MAT-21 MCQ 12, 16). <figure> {{fig:F08_dirichlet_tags}} <figcaption>What to notice: On the exact same partition of [0,1], choosing rational tags puts every rectangle at height 1 (S=1), while choosing irrational tags puts every rectangle at height 0 (S=0). What it does not prove: One partition is not enough; because rationals and irrationals are dense in every subinterval no matter how small \eta > 0 is, the gap \lvert 1 - 0\rvert = 1 \ge 1/2 never shrinks.</figcaption> </figure> Negation of Cauchy Criterion (Bartle Ex 7.2.1) & Example 7.2.2(b) C7 Negation of 7.2.1: f

\notin \mathcal{R}[a,b] if and only if there exists \varepsilon_0 > 0 such that for every \eta > 0 (or for every n \in \mathbb{N} with \eta = 1/n) there exist tagged partitions \dot P and \dot Q with \lVert P\rVert < \eta and \lVert Q\rVert < \eta such that \lvert S(f;\dot P) - S(f;\dot Q)\rvert \ge \varepsilon_0. Example 7.2.2(b) (Dirichlet's function): Let f(x) := 1 if x \in [0,1] is rational and f(x) := 0 if x \in [0,1] is irrational. Then f \notin \mathcal{R}[0,1]. In plain words: to disprove integrability, pick one fixed positive gap \varepsilon_0 (like 1/2) and show that no matter how fine the mesh \eta is, you can still find two taggings \dot P, \dot Q whose sums differ by at least \varepsilon_0. Skeleton · prove f \notin \mathcal{R}[a,b] via Cauchy CriterionTake \varepsilon_0 = $? $ > 0. Given any \eta > 0, choose a partition P of [a,b] with \lVert P\rVert < \eta. Form \dot P by choosing rational tags t_i \in I_i, giving $S(f;\dot P) = $?, and form \dot Q by choosing irrational tags s_i \in I_i, giving $S(f;\dot Q) = $?. Then \lvert S(f;\dot P) - S(f;\dot Q)\rvert \ge \varepsilon_0. By the Cauchy Criterion (7.2.1), f \notin \mathcal{R}[a,b]. ∎ Proof that Dirichlet's function is not in \mathcal{R}[0,1] (Bartle p. 209) Take \varepsilon_0 := \tfrac12 > 0, and let \eta > 0 be arbitrary. Choose any partition P = \{[x_{i-1},x_i]\}_{i=1}^n of [0,1] with \lVert P\rVert < \eta (for instance, an equal partition with n > 1/\eta).Set the fixed gap \varepsilon_0 = 1/2 and let the opponent hand us any \eta > 0. By the density of \mathbb{Q} in \mathbb{R}, each subinterval [x_{i-1},x_i] contains a rational tag t_i. Let \dot P be P with these rational tags; then f(t_i) = 1 for all i, so
S(f;\dot P) = \sum_{i=1}^n 1\cdot(x_i - x_{i-1}) = 1 - 0 = 1.
Density of rationals (P14) and telescoping sum (M01). By the density of \mathbb{R}\setminus\mathbb{Q} in \mathbb{R}, each subinterval [x_{i-1},x_i] contains an irrational tag s_i. Let \dot Q be P with these irrational tags; then f(s_i) = 0 for all i, so S(f;\dot Q) = 0.Density of irrationals (P14). Both \lVert P\rVert < \eta and \lVert Q\rVert < \eta, yet \lvert S(f;\dot P) - S(f;\dot Q)\rvert = \lvert 1 - 0\rvert = 1 \ge \tfrac12 = \varepsilon_0. Thus the Cauchy Criterion 7.2.1 fails, and f \notin \mathcal{R}[0,1]. ∎Negation of Theorem 7.2.1.   Worked example · h(x) = x + 1 on rationals, 0 on irrationals (Bartle Ex 7.2.2; PYQ MAT-21 MCQ 16) Let h(x) := x + 1 for x \in [0,1] \cap \mathbb{Q} and h(x) := 0 for x \in [0,1] \setminus \mathbb{Q}. Take \varepsilon_0 := 1. For any \eta > 0, choose a partition P of [0,1] with \lVert P\rVert < \eta. If \dot P has all rational tags t_i \in [x_{i-1},x_i], then h(t_i) = t_i + 1 \ge 1 on [0,1], so S(h;\dot P) = \sum_{i=1}^n (t_i + 1)(x_i - x_{i-1}) \ge \sum_{i=1}^n 1\cdot(x_i - x_{i-1}) = 1. If \dot Q has all irrational tags, then S(h;\dot Q) = 0. Hence \lvert S(h;\dot P) - S(h;\dot Q)\rvert \ge 1 = \varepsilon_0, so h \notin \mathcal{R}[0,1]. Your intuition will lie to you here "There are vastly more irrationals than rationals, so Dirichlet's function is 'almost always 0' and its integral should be 0." That is true for the Lebesgue integral (taught in postgraduate Measure Theory), which gives 0. It is false for the Riemann integral, because Riemann sums let an opponent pick rational tags in every subinterval! In the model paper: Q24 (PYQ MAT-22 Q4, MAT-22 MCQ 17, MAT-21 MCQ 12, 16). </section>
<!-- T12 (K12) --> <section class="page"> T12<h2>The Squeeze Theorem for integrals</h2> Proving f \in \mathcal{R}[a,b] by trapping f between two simpler integrable

functions \alpha_\varepsilon \le f \le \omega_\varepsilon (Bartle Theorem 7.2.3, Example 7.2.6(b); PYQ MTS-26 Q18, MTS-22 Q6, MAT-21 Q7). <figure> {{fig:F05_squeeze_step}} <figcaption>What to notice: On [0,1] with n=4, the line h(x)=x is trapped between the lower step function \alpha_4 (integral 3/8) and upper step function \omega_4 (integral 5/8); the shaded gap consists of 4 squares of area (1/4)^2=1/16, giving \int_0^1(\omega_4-\alpha_4)=1/4=1/n. What it does not prove: One picture for n=4 does not prove h\in\mathcal{R}[0,1]; the Squeeze Theorem requires \int_0^1(\omega_n-\alpha_n) = 1/n < \varepsilon for every \varepsilon>0 by choosing n > 1/\varepsilon.</figcaption> </figure> Theorem 7.2.3 · Squeeze Theorem (Bartle p. 209) · say it word for wordLet f : [a,b] \to \mathbb{R}. Then f \in \mathcal{R}[a,b] if and only if for every \varepsilon > 0 there exist functions \alpha_\varepsilon and \omega_\varepsilon in \mathcal{R}[a,b] with
\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) \quad \text{for all } x \in [a,b]
and such that
\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon.
In plain words: instead of wrestling with all possible tags of f, trap f between a floor function \alpha_\varepsilon \in \mathcal{R}[a,b] and a ceiling function \omega_\varepsilon \in \mathcal{R}[a,b] whose integrals are within \varepsilon of each other. Hypotheses · Theorem 7.2.3 \alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b] with \alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) for all x \in [a,b]if \alpha_\varepsilon, \omega_\varepsilon are not themselves integrable, trapping f between them tells us nothing \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon holds for every \varepsilon > 0 (so \alpha_\varepsilon, \omega_\varepsilon must be adjusted as \varepsilon shrinks)Bartle Ex 7.2.4 trap: one fixed pair like \alpha(x) = -x and \omega(x) = x on [0,1] satisfies -x \le f(x) \le x, but \int_0^1 (\omega - \alpha) = \int_0^1 2x\,dx = 1, which is NOT < \varepsilon for \varepsilon \le 1. Indeed, f(x) = x on \mathbb{Q} and -x on irrationals satisfies -x \le f(x) \le x yet f \notin \mathcal{R}[0,1]! Skeleton · prove f \in \mathcal{R}[a,b] using the Squeeze TheoremLet \varepsilon > 0 be given. Define \alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b] by ?. Then \alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) on [a,b], and \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) = $? $ < \varepsilon. By the Squeeze Theorem (7.2.3), f \in \mathcal{R}[a,b]. ∎ </section>
<section class="page"> Know(\Leftarrow) For every \varepsilon > 0, \alpha_\varepsilon \le f \le \omega_\varepsilon with \alpha_\varepsilon,\omega_\varepsilon \in \mathcal{R}[a,b] and \int_a^b(\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon.WantShow \lvert S(f;\dot P) - S(f;\dot Q)\rvert < 3\varepsilon when \lVert P\rVert, \lVert Q\rVert < \delta_\varepsilon, and invoke the Cauchy Criterion 7.2.1. Proof of Theorem 7.2.3 (Bartle pp. 209–210) (\Rightarrow) If f \in \mathcal{R}[a,b], simply take \alpha_\varepsilon := f and \omega_\varepsilon := f for all \varepsilon > 0; then \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) = \int_a^b 0 = 0 < \varepsilon.Trivial choice when f is already known to be integrable. (\Leftarrow) Let \varepsilon > 0. Since \alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b], there exists \delta_\varepsilon > 0 such that if \dot P is any tagged partition with \lVert P\rVert < \delta_\varepsilon, then

\left\lvert S(\alpha_\varepsilon;\dot P) - \int_a^b \alpha_\varepsilon\right\rvert < \varepsilon \quad \text{and} \quad \left\lvert S(\omega_\varepsilon;\dot P) - \int_a^b \omega_\varepsilon\right\rvert < \varepsilon.
Definition 7.1.1 applied to \alpha_\varepsilon and \omega_\varepsilon, taking the minimum of the two deltas (M08). Unpacking these two inequalities gives \int_a^b \alpha_\varepsilon - \varepsilon < S(\alpha_\varepsilon;\dot P) and S(\omega_\varepsilon;\dot P) < \int_a^b \omega_\varepsilon + \varepsilon. Since \alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) on [a,b], we have S(\alpha_\varepsilon;\dot P) \le S(f;\dot P) \le S(\omega_\varepsilon;\dot P), and therefore \int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot P) < \int_a^b \omega_\varepsilon + \varepsilon.
Pointwise inequality \alpha_\varepsilon \le f \le \omega_\varepsilon preserves Riemann sums (P02). If \dot Q is any other tagged partition with \lVert Q\rVert < \delta_\varepsilon, then the same bounds hold for S(f;\dot Q): \int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot Q) < \int_a^b \omega_\varepsilon + \varepsilon.
Step 2 applied to \dot Q. Subtracting the two inequalities and using linearity (Theorem 7.1.5) together with \int_a^b(\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon gives
\lvert S(f;\dot P) - S(f;\dot Q)\rvert < \int_a^b \omega_\varepsilon - \int_a^b \alpha_\varepsilon + 2\varepsilon = \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) + 2\varepsilon < \varepsilon + 2\varepsilon = 3\varepsilon.
Two numbers in (A-\varepsilon, B+\varepsilon) are within (B-A)+2\varepsilon of each other; Theorem 7.1.5(a),(b). Since \varepsilon > 0 is arbitrary, the Cauchy Criterion (Theorem 7.2.1) implies that f \in \mathcal{R}[a,b]. ∎Theorem 7.2.1 (replacing \varepsilon by \varepsilon/3 if desired).   Worked example · Squeeze applied to h(x)=x on [0,1] (Bartle Example 7.2.6(b)) Partition [0,1] into n equal pieces using P_n = (0, \tfrac1n, \tfrac2n, \dots, \tfrac{n-1}{n}, 1). On [\tfrac{k-1}{n}, \tfrac{k}{n}) for k=1,\dots,n-1 (and on [\tfrac{n-1}{n},1] for k=n), define step functions \alpha_n(x) := \tfrac{k-1}{n} (the minimum of x on the piece) and \omega_n(x) := \tfrac{k}{n} (the maximum of x on the piece). By Theorem 7.2.5 and P03,

\int_0^1 \alpha_n = \frac{1}{n}\left(0 + \frac{1}{n} + \cdots + \frac{n-1}{n}\right) = \frac{1}{n^2}\cdot\frac{(n-1)n}{2} = \frac{1}{2}\left(1 - \frac{1}{n}\right),

\int_0^1 \omega_n = \frac{1}{n}\left(\frac{1}{n} + \frac{2}{n} + \cdots + \frac{n}{n}\right) = \frac{1}{n^2}\cdot\frac{n(n+1)}{2} = \frac{1}{2}\left(1 + \frac{1}{n}\right).
 Since \alpha_n \le h \le \omega_n and \int_0^1(\omega_n - \alpha_n) = \tfrac1n < \varepsilon for any n > 1/\varepsilon, Theorem 7.2.3 gives h \in \mathcal{R}[0,1]. Moreover, \tfrac12(1 - \tfrac1n) \le \int_0^1 h \le \tfrac12(1 + \tfrac1n) for all n \in \mathbb{N} forces \int_0^1 x\,dx = \tfrac12. In the model paper: Q25 (PYQ MTS-26 Q18, MTS-22 Q6, MAT-21 Q7). </section>
<!-- T13 (K13) & T14 (K14) --> <section class="page"> T13<h2>Elementary step functions \varphi_J</h2> A single flat block of height 1 on a subinterval J with endpoints c < d has integral d - c (Bartle Lemma 7.2.4, Ex 7.1.13; PYQ MAT-21 MCQ 10). Lemma 7.2.4 · Elementary Step Functions (Bartle p. 210) · say it word

for wordIf J is a subinterval of $[a,b]$ having endpoints $c < d$, and if $\varphi_J(x) := 1$ for $x \in J$ and $\varphi_J(x) := 0$ elsewhere on $[a,b]$, then $\varphi_J \in \mathcal{R}[a,b]$ and $\int_a^b \varphi_J = d - c$. In plain words: whether the subinterval J is closed $[c,d]$, open (c,d), or half-open $[c,d)$ or $(c,d]$, the indicator function φ_J is integrable and its integral is simply the length $d - c$ of J. Skeleton · prove Lemma 7.2.4First let $J = [c,d]$. For any $\dot P$ with $\lVert P\rVert < \delta$, the union of subintervals with tag in $[c,d]$ contains $[c+\delta,d-\delta]$ and lies inside $[c-\delta,d+\delta]$, so $(d-c) - 2\delta \le S(\varphi_{[c,d]};\dot P) \le (d-c) + 2\delta$. Choose $\delta_\varepsilon = \varepsilon/4$ to get $\lvert S - (d-c)\rvert < \varepsilon$. The other three intervals (c,d), $[c,d)$, $(c,d]$ differ from $[c,d]$ at at most 2 endpoints, so Theorem 7.1.3 gives the same integral $d - c$. ∎ Proof of Lemma 7.2.4 & Exercise 7.1.13 (Bartle p. 210) First suppose $J = [c,d]$ with $a \le c < d \le b$. Let $\dot P$ be any tagged partition of $[a,b]$ with $\lVert P\rVert < \delta$, and let $\dot P_1$ be the subset of subintervals whose tags belong to $[c,d]$.Outside $\dot P_1$, $\varphi_{[c,d]}(t_i) = 0$, so $S(\varphi_{[c,d]};\dot P) = S(\varphi_{[c,d]};\dot P_1)$. By M10 (Exercise 7.1.5), the union U_1 of the subintervals in $\dot P_1$ satisfies $[c+\delta,\, d-\delta] \subseteq U_1 \subseteq [c-\delta,\, d+\delta]$. Since $\varphi_{[c,d]}(t_i) = 1$ on $\dot P_1$, we have
$$(d - c) - 2\delta \le S(\varphi_{[c,d]};\dot P) \le (d - c) + 2\delta, \quad \text{so} \quad \lvert S(\varphi_{[c,d]};\dot P) - (d - c)\rvert \le 2\delta.$$
Length of $[c+\delta,d-\delta]$ is $(d-c)-2\delta$ and length of $[c-\delta,d+\delta]$ is $(d-c)+2\delta$. Given $\varepsilon > 0$, choose $\delta_\varepsilon := \varepsilon/4 > 0$; then $\lVert P\rVert < \delta_\varepsilon \Rightarrow \lvert S(\varphi_{[c,d]};\dot P) - (d - c)\rvert \le 2(\varepsilon/4) = \varepsilon/2 < \varepsilon$. Hence $\varphi_{[c,d]} \in \mathcal{R}[a,b]$ and $\int_a^b \varphi_{[c,d]} = d - c$.Definition 7.1.1 with $L = d - c$. If J is (c,d), $[c,d)$, or $(c,d]$, then φ_J differs from $\varphi_{[c,d]}$ only at one or both of the endpoints c, d (at most two points). By Theorem 7.1.3, $\varphi_J \in \mathcal{R}[a,b]$ and $\int_a^b \varphi_J = \int_a^b \varphi_{[c,d]} = d - c$. ∎Changing a function at finitely many points (Theorem 7.1.3, T06).   Check yourself · PYQ MAT-21 MCQ 10 Let $g : [0,4] \to \mathbb{R}$ be $g(x) = 3$ for $0 \le x \le 2$ and $g(x) = 2$ for $2 < x \le 4$. Express g using elementary step functions and find $\int_0^4 g$. $g = 3\varphi_{[0,2]} + 2\varphi_{(2,4]}$. By Lemma 7.2.4 and linearity (7.1.5), $\int_0^4 g = 3(2 - 0) + 2(4 - 2) = 6 + 4 = 10$. In the model paper: Q8 (PYQ MAT-21 MCQ 10). </section>

<section class="page"> T14<h2>Step functions are integrable</h2> Every step function is a linear combination of elementary step functions φ_{J_j} (Bartle Theorem 7.2.5, Example 7.2.6(a); PYQ MTS-26 Q5, MAT-22 Q16). <figure> {{fig:F16_elementary_steps}} <figcaption>What to notice: The step function g of Example 7.1.4(b) splits into $2\varphi_{[0,1]}$ (area $2\times 1=2$) plus $3\varphi_{(1,3]}$ (area $3\times 2=6$), so $\int_0^3 g = 2 + 6 = 8$ without touching ε or δ. What it does not prove: This formula only works for a finite sum of interval indicators; a function that jumps infinitely often is not a step function.</figcaption> </figure> Theorem 7.2.5 · Integrability of Step Functions (Bartle p. 210, eqs. (4)–(5))If $\varphi : [a,b] \to \mathbb{R}$ is a step function, then $\varphi \in \mathcal{R}[a,b]$. Moreover, if $\varphi = \sum_{j=1}^m k_j\varphi_{J_j}$ where J_j has endpoints $c_j < d_j$, then

\int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j).
In plain words: every step function is integrable, and its integral is the sum of (constant height k_j) \times (interval length d_j - c_j). Hypotheses · Theorem 7.2.5 \varphi is a step function on [a,b]: [a,b] is the union of finitely many non-overlapping intervals J_1,\dots,J_m on each of which \varphi is constant (Definition 5.4.9)Bartle Ex 7.2.6 non-example: taking only finitely many values is NOT enough to be a step function! Dirichlet's function C7 takes only two values (0 and 1), yet the sets \mathbb{Q}\cap[0,1] and [0,1]\setminus\mathbb{Q} are not finite unions of intervals, and f \notin \mathcal{R}[0,1]. Proof of Theorem 7.2.5 (Bartle pp. 210–211) By Definition 5.4.9 (and Exercise 7.2.5), [a,b] is the union of finitely many non-overlapping subintervals J_1, \dots, J_m with endpoints c_j \le d_j such that \varphi(x) = k_j on J_j. Disjointness of the pieces (up to single endpoints, which can be absorbed into one piece or treated as single-point intervals) allows us to write
\varphi = \sum_{j=1}^m k_j\varphi_{J_j},
where each \varphi_{J_j} is an elementary step function on [a,b].Decomposition into indicator functions (M06). By Lemma 7.2.4, each \varphi_{J_j} belongs to \mathcal{R}[a,b] and satisfies \int_a^b \varphi_{J_j} = d_j - c_j.Lemma 7.2.4 (T13). By Theorem 7.1.5(a) and (b) (applied inductively to the finite sum of m terms, Exercise 7.1.7), the linear combination \varphi = \sum_{j=1}^m k_j\varphi_{J_j} belongs to \mathcal{R}[a,b] and \int_a^b \varphi = \sum_{j=1}^m k_j\int_a^b \varphi_{J_j} = \sum_{j=1}^m k_j(d_j - c_j). \quad \blacksquare
Linearity of the integral (Theorem 7.1.5, T07).   Worked examples · Bartle Example 7.2.6(a) & Exercise 7.2.7 1. Example 7.2.6(a): The function g(x) = 2 on [0,1] and g(x) = 3 on (1,3] of Example 7.1.4(b) is the step function 2\varphi_{[0,1]} + 3\varphi_{(1,3]}, so Theorem 7.2.5 immediately gives \int_0^3 g = 2(1 - 0) + 3(3 - 1) = 8 in one line! 2. Every Riemann sum is the integral of a step function (Bartle Ex 7.2.7): Given any f : [a,b] \to \mathbb{R} and any tagged partition \dot P = \{([x_{i-1},x_i],t_i)\}_{i=1}^n, define the step function \varphi(x) := f(t_i) on [x_{i-1},x_i) for i=1,\dots,n-1 and \varphi(x) := f(t_n) on [x_{n-1},x_n]. By Theorem 7.2.5, \int_a^b \varphi = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) = S(f;\dot P). In the model paper: Q26 (PYQ MTS-26 Q5, MAT-22 Q16). </section>
<!-- T15 (K15) --> <section class="page"> T15<h2>Continuous functions on [a,b] are integrable</h2> Uniform Continuity + Max–Min + Step functions + Squeeze = Theorem 7.2.7 (PYQ MTS-25 Q18, MAT-21 Q30, MTS-22 Q19, MAT-22 MCQ 21, 23, MAT-21 MCQ 15). <figure> {{fig:F06_cont_min_max_steps}} <figcaption>What to notice: On each subinterval I_i, the continuous function f(x)=\sin x attains its minimum at u_i (floor step \alpha_\varepsilon) and maximum at v_i (ceiling step \omega_\varepsilon); uniform continuity makes every vertical gap f(v_i)-f(u_i) < \varepsilon/(b-a) once \lVert P\rVert < \delta_\varepsilon. What it does not prove: Continuity on an open interval (a,b) is not enough (1/x on (0,1] is not integrable).</figcaption> </figure> Theorem 7.2.7 · Integrability of Continuous Functions (Bartle p. 212) · say it word for wordIf f : [a,b] \to \mathbb{R} is continuous on [a,b], then f \in \mathcal{R}[a,b]. In plain words: every continuous function on a closed bounded interval is Riemann integrable.

Hypotheses & Converse · Theorem 7.2.7 f is continuous on [a,b]Dirichlet C7 is defined on [0,1] and bounded, but discontinuous everywhere and \notin \mathcal{R}[0,1] the interval [a,b] is closed and boundedf(x) = 1/x on (0,1] with f(0) = 0 C16 is continuous on the half-open interval (0,1], yet unbounded and \notin \mathcal{R}[0,1] Converse is FALSE (PYQ MAT-22 MCQ 21): f \in \mathcal{R}[a,b] does not imply f is continuous or uniformly continuous on [a,b]the step function g of 7.1.4(b), \operatorname{sgn} C9, and Thomae's function C8 are all in \mathcal{R}[a,b] but discontinuous (and hence not uniformly continuous) Knowf is continuous on the closed bounded interval [a,b]; \varepsilon > 0 is given.WantConstruct step functions \alpha_\varepsilon \le f \le \omega_\varepsilon on [a,b] with \int_a^b(\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon, then apply the Squeeze Theorem 7.2.3. Scratch · why we need Uniform Continuity (5.4.3) AND Max–Min (5.3.4) On I_i = [x_{i-1},x_i], we want step heights \alpha_\varepsilon(x) = \min_{I_i} f = f(u_i) and \omega_\varepsilon(x) = \max_{I_i} f = f(v_i). How do we know u_i, v_i exist? Because f is continuous on the closed bounded interval I_i (Max–Min Theorem 5.3.4). Then \int_a^b(\omega_\varepsilon - \alpha_\varepsilon) = \sum_{i=1}^n (f(v_i) - f(u_i))(x_i - x_{i-1}). To make this sum < \varepsilon, we need f(v_i) - f(u_i) < \varepsilon/(b-a) on every subinterval at once. Since \lvert v_i - u_i\rvert \le x_i - x_{i-1} \le \lVert P\rVert, the Uniform Continuity Theorem 5.4.3 gives one \delta_\varepsilon > 0 that works across all i=1,\dots,n simultaneously! Proof of Theorem 7.2.7 (Bartle p. 212) By the Uniform Continuity Theorem 5.4.3 (P08), since f is continuous on the closed bounded interval [a,b], f is uniformly continuous on [a,b]. Therefore, given any \varepsilon > 0, there exists \delta_\varepsilon > 0 such that if u, v \in [a,b] and \lvert u - v\rvert < \delta_\varepsilon, then \lvert f(u) - f(v)\rvert < \frac{\varepsilon}{b - a}.
Uniform Continuity Theorem 5.4.3 applied with tolerance \varepsilon/(b-a) > 0. Let P = \{I_i\}_{i=1}^n be a partition of [a,b] with \lVert P\rVert < \delta_\varepsilon, where I_i = [x_{i-1},x_i].Choose an equal partition with n > (b-a)/\delta_\varepsilon so every subinterval has length <\delta_\varepsilon. Since f is continuous on each closed bounded subinterval I_i, the Maximum–Minimum Theorem 5.3.4 (P07) implies that there exist points u_i, v_i \in I_i where f attains its minimum and maximum values on I_i, respectively: f(u_i) \le f(x) \le f(v_i) for all x \in I_i.Maximum–Minimum Theorem 5.3.4 on each I_i = [x_{i-1},x_i]. Define the step function \alpha_\varepsilon : [a,b] \to \mathbb{R} by \alpha_\varepsilon(x) := f(u_i) for x \in [x_{i-1},x_i) (i=1,\dots,n-1) and \alpha_\varepsilon(x) := f(u_n) for x \in [x_{n-1},x_n]. Similarly, define \omega_\varepsilon(x) := f(v_i) for x \in [x_{i-1},x_i) (i=1,\dots,n-1) and \omega_\varepsilon(x) := f(v_n) for x \in [x_{n-1},x_n]. Then \alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b] by Theorem 7.2.5, and
\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) \quad \text{for all } x \in [a,b].
Construction of disjoint half-open intervals and Theorem 7.2.5 (T14). Since u_i, v_i \in I_i and \lVert P\rVert < \delta_\varepsilon, we have \lvert u_i - v_i\rvert \le x_i - x_{i-1} < \delta_\varepsilon, so f(v_i) - f(u_i) < \varepsilon/(b-a) by Step 1. Using formula (5) of Theorem 7.2.5,

0 \le \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) = \sum_{i=1}^n \big(f(v_i) - f(u_i)\big)(x_i - x_{i-1}) < \sum_{i=1}^n \frac{\varepsilon}{b-a}(x_i - x_{i-1}) = \frac{\varepsilon}{b-a}(b-a) = \varepsilon.
Step-function integral formula (T14), Step 1 bound, and telescoping \sum(x_i - x_{i-1}) = b-a (M01). Since \varepsilon > 0 is arbitrary, the Squeeze Theorem 7.2.3 implies that f \in \mathcal{R}[a,b]. ∎Squeeze Theorem 7.2.3 (T12).   </section>
<section class="page"> Where each hypothesis was used & Two classic exam applications Where hypotheses were used: Continuity of f and closedness/boundedness of [a,b] were used in Step 1 (Uniform Continuity Theorem 5.4.3) and in Step 3 (Maximum–Minimum Theorem 5.3.4 on each I_i). Application 1 · f \ge 0 continuous and \int_a^b f = 0 \Rightarrow f \equiv 0 (Bartle Ex 7.2.8–9; PYQ MAT-22 MCQ 23, Q18): Suppose for contradiction that f(c) > 0 for some c \in [a,b]. By continuity at c with \varepsilon = \tfrac12 f(c) > 0, there is a subinterval J = [c_1, d_1] \subseteq [a,b] of length d_1 - c_1 > 0 containing c on which f(x) > \tfrac12 f(c). Then the elementary step function satisfies 0 \le \tfrac12 f(c)\varphi_J(x) \le f(x) on [a,b], so by Theorem 7.1.5(c) and Lemma 7.2.4, \int_a^b f \ge \tfrac12 f(c)(d_1 - c_1) > 0, contradicting \int_a^b f = 0! Hence f(x) = 0 for all x \in [a,b]. (Continuity cannot be dropped: f(0)=1 and f(x)=0 on (0,1] has f \ge 0 and \int_0^1 f = 0, yet f(0) \ne 0.) Application 2 · Mean Value Theorem for Integrals (Bartle Ex 7.2.16, Q20): If f is continuous on [a,b] (a < b), then by Max–Min (5.3.4) f attains its minimum m and maximum M on [a,b]. Since m \le f(x) \le M, Theorem 7.1.5(c) gives m(b-a) \le \int_a^b f \le M(b-a), so k := \frac{1}{b-a}\int_a^b f \in [m,M]. By Bolzano's Intermediate Value Theorem 5.3.7, there exists c \in [a,b] with f(c) = k, i.e. \int_a^b f = f(c)(b-a). In the model paper: Q18, Q20, Q37 (PYQ MTS-25 Q18, MAT-21 Q30, MTS-22 Q19, MAT-22 MCQ 21, 23, MAT-21 MCQ 15). </section>
<!-- T16 (K16) --> <section class="page"> T16<h2>Monotone functions on [a,b] are integrable</h2> Even with countably many jumps, an increasing or decreasing function on [a,b] is integrable because the staircase gap telescopes (Bartle Theorem 7.2.8; PYQ MTS-26 Q19, MAT-22 Q30, MAT-22 MCQ 22). <figure> {{fig:F07_monotone_staircase}} <figcaption>What to notice: For the increasing function f(x)=x^2 on [0,1] with n=4 equal pieces, sliding all four error rectangles (\omega-\alpha) horizontally stacks them into a single column of width (b-a)/n = 1/4 and height f(1)-f(0) = 1, so \int_0^1(\omega-\alpha) = 1/4. What it does not prove: The picture shows n=4; the proof works by choosing n > (b-a)(f(b)-f(a))/\varepsilon for any \varepsilon > 0.</figcaption> </figure> Theorem 7.2.8 · Integrability of Monotone Functions (Bartle p. 212) · say it word for wordIf f : [a,b] \to \mathbb{R} is monotone on [a,b], then f \in \mathcal{R}[a,b]. In plain words: every increasing or decreasing function on a closed bounded interval [a,b] is Riemann integrable. Hypotheses & Converse · Theorem 7.2.8 f is monotone (increasing or decreasing) on the closed bounded interval [a,b]if f is only monotone on the open interval (0,1], e.g. f(x) = 1/x on (0,1] with f(0)=0 C16, f is unbounded and \notin \mathcal{R}[0,1]. On a closed interval [a,b], a monotone function is automatically bounded between f(a) and f(b)! Converse is FALSE (PYQ MAT-22 MCQ 22): f \in \mathcal{R}[a,b] does not

imply f is monotone on [a,b]f(x) = \sin x on [0,\pi] and f(x) = x^2 on [-1,1] are continuous (hence in \mathcal{R}) but not monotone; Thomae's function C8 is not monotone on any subinterval Knowf is increasing on [a,b] (for decreasing f, apply to -f and use Theorem 7.1.5(a)); \varepsilon > 0 is given.WantOn n equal subintervals of length (b-a)/n, define step functions \alpha \le f \le \omega with \int_a^b(\omega-\alpha) = \frac{b-a}{n}(f(b)-f(a)) < \varepsilon. Proof of Theorem 7.2.8 (Bartle p. 212) Assume f is increasing on I = [a,b] (if f is decreasing, then -f is increasing, so -f \in \mathcal{R}[a,b] and hence f = -(-f) \in \mathcal{R}[a,b] by Theorem 7.1.5(a)). For n \in \mathbb{N}, partition I = [a,b] into n equal subintervals I_k = [x_{k-1},x_k] of length x_k - x_{k-1} = (b-a)/n for k=1,\dots,n.Equal widths (b-a)/n allow us to factor out the common width and telescope the heights. Since f is increasing on I_k, its minimum on I_k is attained at the left endpoint x_{k-1} and its maximum at the right endpoint x_k: f(x_{k-1}) \le f(x) \le f(x_k) for all x \in I_k.Definition of increasing function (P10)—no continuity needed! Define step functions \alpha, \omega : [a,b] \to \mathbb{R} by \alpha(x) := f(x_{k-1}) and \omega(x) := f(x_k) for x \in [x_{k-1},x_k) (k=1,\dots,n-1), and \alpha(x) := f(x_{n-1}), \omega(x) := f(x_n) for x \in [x_{n-1},x_n]. Then \alpha, \omega \in \mathcal{R}[a,b] by Theorem 7.2.5 and \alpha(x) \le f(x) \le \omega(x) on [a,b].Step-function integrability (Theorem 7.2.5, T14). By Theorem 7.2.5,
$$\int_a^b \alpha = \frac{b-a}{n}\big(f(x_0) + f(x_1) + \cdots + f(x_{n-1})\big), \quad \int_a^b \omega = \frac{b-a}{n}\big(f(x_1) + \cdots + f(x_{n-1}) + f(x_n)\big).$$
Subtracting and cancelling the interior terms f(x_1),\dots,f(x_{n-1}) gives
$$\int_a^b (\omega - \alpha) = \frac{b-a}{n}\big(f(x_n) - f(x_0)\big) = \frac{b-a}{n}\big(f(b) - f(a)\big).$$
Telescoping sum (M01, M05) with x_0 = a and x_n = b. Given \varepsilon > 0, choose n \in \mathbb{N} such that n > (b-a)(f(b)-f(a))/\varepsilon. Then \int_a^b (\omega - \alpha) < \varepsilon. By the Squeeze Theorem 7.2.3, f \in \mathcal{R}[a,b]. ∎Archimedean property (P13) and Squeeze Theorem 7.2.3 (T12).   Worked example · Staircase for f(x) = x^2 on [0,1] with n=4 With P_4 = (0, \tfrac14, \tfrac12, \tfrac34, 1), the function f(x) = x^2 is increasing on [0,1]. The lower step heights are 0, \tfrac1{16}, \tfrac14, \tfrac9{16} and upper step heights are \tfrac1{16}, \tfrac14, \tfrac9{16}, 1, so:
$$\int_0^1 \alpha = \frac14\left(0 + \frac1{16} + \frac4{16} + \frac9{16}\right) = \frac{14}{64} = 0.21875, \qquad \int_0^1 \omega = \frac14\left(\frac1{16} + \frac4{16} + \frac9{16} + 1\right) = \frac{30}{64} = 0.46875.$$
 Their difference is 0.46875 - 0.21875 = 0.25 = \frac{1-0}{4}(f(1)-f(0)), trapping the exact value \int_0^1 x^2\,dx = \tfrac13 \approx 0.3333. In the model paper: Q38 (PYQ MTS-26 Q19, MAT-22 Q30, MAT-22 MCQ 22). </section>
<!-- T17 (K17) --> <section class="page"> T17<h2>The Additivity Theorem</h2> Splitting an integral at an interior point c \in (a,b) (Bartle Theorem 7.2.9; PYQ MTS-26 Q25, MAT-22 Q37a, MTS-22 Q25). <figure> {{fig:F13_additivity_split}} <figcaption>What to notice: When c=1 falls inside the k-th subinterval [x_{k-1},x_k]=[0.8,1.3] of \dot Q, we split that subinterval at c into ([0.8,1],1) and ([1,1.3],1) with tag c; only that single rectangle changes, by at most 2M(x_k-x_{k-1}) < \varepsilon/3. What it does not prove: You must prove both

directions (\Leftarrow) and (\Rightarrow), and in (\Leftarrow) you must handle Case (ii) where c is not a partition point.</figcaption> </figure> Theorem 7.2.9 · Additivity Theorem (Bartle p. 213, eq. (6)) · say it word for wordLet f : [a,b] \to \mathbb{R} and let c \in (a,b). Then f \in \mathcal{R}[a,b] if and only if its restrictions to [a,c] and to [c,b] are both Riemann integrable. In this case
\int_a^b f = \int_a^c f + \int_c^b f.
In plain words: f is integrable on the whole interval [a,b] if and only if it is integrable on the left piece [a,c] and on the right piece [c,b], and the whole integral is the sum of the two piece integrals. Hypotheses · Theorem 7.2.9 for (\Leftarrow): both restrictions f|_{[a,c]} \in \mathcal{R}[a,c] and f|_{[c,b]} \in \mathcal{R}[c,b]if f=0 on [0,1] and f=\text{Dirichlet} C7 on [1,2], then f|_{[0,1]} \in \mathcal{R}[0,1] but f|_{[1,2]} \notin \mathcal{R}[1,2], so f \notin \mathcal{R}[0,2] boundedness of f on [a,b] in (\Leftarrow)since f|_{[a,c]} \in \mathcal{R}[a,c] and f|_{[c,b]} \in \mathcal{R}[c,b], Theorem 7.1.6 implies each piece is bounded, so f is bounded on [a,b] by some M > 0 (state this in the exam—it carries a mark!) Know(\Leftarrow) f_1 = f|_{[a,c]} \in \mathcal{R}[a,c] with integral L_1, f_2 = f|_{[c,b]} \in \mathcal{R}[c,b] with integral L_2. (\Rightarrow) f \in \mathcal{R}[a,b].Want(\Leftarrow) Show \lvert S(f;\dot Q) - (L_1+L_2)\rvert < \varepsilon for \lVert Q\rVert < \delta_\varepsilon. (\Rightarrow) Use Cauchy 7.2.1 by adjoining the same tagged partition of [c,b] to \dot P_1, \dot Q_1. Scratch · the three \varepsilon/3 errors in (\Leftarrow) Case (ii) When c \in (x_{k-1},x_k) is not a partition point of \dot Q, split I_k at c with tag c on both halves to get \dot Q_1 on [a,c] and \dot Q_2 on [c,b]. All other intervals match! So S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2) = f(t_k)(x_k-x_{k-1}) - f(c)(c-x_{k-1}) - f(c)(x_k-c) = (f(t_k) - f(c))(x_k - x_{k-1}). Bound this by 2M\lVert Q\rVert < \varepsilon/3 by choosing \delta_\varepsilon = \min\{\delta', \delta'', \varepsilon/(6M)\}. </section>
<section class="page"> Proof of Theorem 7.2.9 (Bartle pp. 213–214) (\Leftarrow) Suppose f_1 := f|_{[a,c]} \in \mathcal{R}[a,c] with integral L_1 and f_2 := f|_{[c,b]} \in \mathcal{R}[c,b] with integral L_2. Given \varepsilon > 0, choose \delta' > 0 such that \lVert P_1\rVert < \delta' \Rightarrow \lvert S(f_1;\dot P_1) - L_1\rvert < \varepsilon/3, and choose \delta'' > 0 such that \lVert P_2\rVert < \delta'' \Rightarrow \lvert S(f_2;\dot P_2) - L_2\rvert < \varepsilon/3.Definition 7.1.1 applied to f_1 and f_2 with tolerance \varepsilon/3. By Theorem 7.1.6, f_1 is bounded on [a,c] and f_2 is bounded on [c,b], so f is bounded on [a,b]. Choose M > 0 such that \lvert f(x)\rvert \le M for all x \in [a,b] (if f \equiv 0, take M = 1), and define
\delta_\varepsilon := \min\left\{\delta',\, \delta'',\, \frac{\varepsilon}{6M}\right\} > 0.
Let \dot Q = \{(I_i,t_i)\}_{i=1}^m be any tagged partition of [a,b] with \lVert Q\rVert < \delta_\varepsilon.Boundedness Theorem 7.1.6 on each piece; three-way minimum (M08). Case (i): c is a partition point of \dot Q. Split \dot Q at c into a tagged partition \dot Q_1 of [a,c] and a tagged partition \dot Q_2 of [c,b]. Since S(f;\dot Q) = S(f_1;\dot Q_1) + S(f_2;\dot Q_2) and \lVert Q_1\rVert < \delta', \lVert Q_2\rVert < \delta'', the Triangle Inequality and Step 1 give
\lvert S(f;\dot Q) - (L_1 + L_2)\rvert \le \lvert S(f_1;\dot Q_1) - L_1\rvert + \lvert S(f_2;\dot Q_2) - L_2\rvert < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon.
When c is already a cut point, no rectangle straddles c. Case (ii): c is not a partition point of \dot Q. Then c \in (x_{k-1},x_k) for

some $k \le m$. Define tagged partitions $\dot Q_1$ of $[a,c]$ and $\dot Q_2$ of $[c,b]$ by $\dot Q_1 := \{(I_1,t_1),\dots,(I_{k-1},t_{k-1}),([x_{k-1},c],c)\}$, $\quad \dot Q_2 := \{([c,x_k],c),(I_{k+1},t_{k+1}),\dots,(I_m,t_m)\}$.
Cut the straddling interval $[x_{k-1},x_k]$ at c, using c as the tag for both pieces. Since $(c - x_{k-1}) + (x_k - c) = x_k - x_{k-1}$, all terms of $S(f;\dot Q)$ and $S(f_1;\dot Q_1) + S(f_2;\dot Q_2)$ cancel except the k-th term, giving $S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2) = \big(f(t_k) - f(c)\big)(x_k - x_{k-1})$, and therefore $\lvert S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2)\rvert \le (\lvert f(t_k)\rvert + \lvert f(c)\rvert)(x_k - x_{k-1}) \le 2M\lVert Q\rVert < 2M\cdot\frac{\varepsilon}{6M} = \frac{\varepsilon}{3}$.Triangle Inequality, $\lvert f\rvert \le M$, and $\lVert Q\rVert < \varepsilon/(6M)$. Since $\lVert Q_1\rVert \le \lVert Q\rVert < \delta'$ and $\lVert Q_2\rVert \le \lVert Q\rVert < \delta''$, Step 1 gives $\lvert S(f_1;\dot Q_1) - L_1\rvert < \varepsilon/3$ and $\lvert S(f_2;\dot Q_2) - L_2\rvert < \varepsilon/3$. By the Triangle Inequality,
$\lvert S(f;\dot Q) - (L_1 + L_2)\rvert \le \lvert S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2)\rvert + \lvert S(f_1;\dot Q_1) - L_1\rvert + \lvert S(f_2;\dot Q_2) - L_2\rvert < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$.
Since $\varepsilon > 0$ is arbitrary, $f \in \mathcal{R}[a,b]$ and $\int_a^b f = L_1 + L_2 = \int_a^c f + \int_c^b f$.Three-part Triangle Inequality and Definition 7.1.1 (uniqueness 7.1.2 identifies the integral as $L_1 + L_2$). ($\Rightarrow$) Suppose $f \in \mathcal{R}[a,b]$ and let $\varepsilon > 0$ be given. By the Cauchy Criterion 7.2.1, there exists $\eta_\varepsilon > 0$ such that tagged partitions $\dot P$, $\dot Q$ of $[a,b]$ with norms $<\eta_\varepsilon$ satisfy $\lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon$.Cauchy Criterion 7.2.1 ($\Rightarrow$) applied to f on $[a,b]$. Let $\dot P_1$, $\dot Q_1$ be any tagged partitions of $[a,c]$ with $\lVert P_1\rVert < \eta_\varepsilon$ and $\lVert Q_1\rVert < \eta_\varepsilon$. Choose a single tagged partition $\dot R$ of $[c,b]$ with $\lVert R\rVert < \eta_\varepsilon$, and adjoin $\dot R$ to $\dot P_1$ and $\dot Q_1$ to obtain tagged partitions $\dot P = \dot P_1 \cup \dot R$ and $\dot Q = \dot Q_1 \cup \dot R$ of $[a,b]$ with $\lVert P\rVert < \eta_\varepsilon$ and $\lVert Q\rVert < \eta_\varepsilon$.Using the exact same $\dot R$ on $[c,b]$ makes the $[c,b]$ sums cancel when we subtract! Then $S(f_1;\dot P_1) - S(f_1;\dot Q_1) = \big(S(f_1;\dot P_1) + S(f;\dot R)\big) - \big(S(f_1;\dot Q_1) + S(f;\dot R)\big) = S(f;\dot P) - S(f;\dot Q)$, so
$\lvert S(f_1;\dot P_1) - S(f_1;\dot Q_1)\rvert = \lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon$.
By the Cauchy Criterion 7.2.1, $f_1 \in \mathcal{R}[a,c]$. By the same argument, $f_2 \in \mathcal{R}[c,b]$. Finally, the formula $\int_a^b f = \int_a^c f + \int_c^b f$ follows from the ($\Leftarrow$) part already proved. ∎Cauchy Criterion 7.2.1 ($\Leftarrow$) on $[a,c]$ and $[c,b]$, then apply ($\Leftarrow$) for the equality.   In the model paper: Q39 (PYQ MTS-26 Q25, MAT-22 Q37a, MTS-22 Q25). </section>
<!-- T18 (K18) --> <section class="page"> T18<h2>Restriction and subdivision corollaries</h2> Restricting $f \in \mathcal{R}[a,b]$ to $[c,d] \subseteq [a,b]$ and splitting $[a,b]$ at m points (Bartle Corollaries 7.2.10–7.2.11, Exercises 7.2.11–7.2.13; PYQ MAT-22 Q37b, MAT-22 MCQ 24, 25, MAT-21 MCQ 18). Corollary 7.2.10 & Corollary 7.2.11 (Bartle p. 214) · say them word for word Corollary 7.2.10: If $f \in \mathcal{R}[a,b]$ and if $[c,d] \subseteq [a,b]$, then the restriction of f to $[c,d]$ is in $\mathcal{R}[c,d]$. Corollary 7.2.11: If $f \in \mathcal{R}[a,b]$ and if $a = c_0 < c_1 <$

\cdots < c_m = b, then the restrictions of f to each subinterval [c_{i-1},c_i] are Riemann integrable and
\int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f.
 In plain words: integrability is inherited by every closed subinterval [c,d] \subseteq [a,b], and an integral splits across any finite chain of cut points c_0 < c_1 < \cdots < c_m. Proofs of Corollaries 7.2.10 and 7.2.11 (Bartle p. 214) Proof of 7.2.10 (PYQ MAT-22 Q37b): Let f \in \mathcal{R}[a,b] and a \le c < d \le b. If c > a, then c \in (a,b), so Theorem 7.2.9 (\Rightarrow) implies that the restriction f|_{[c,b]} belongs to \mathcal{R}[c,b] (if c = a, this holds by hypothesis). Next, if d < b, then d \in (c,b), so applying Theorem 7.2.9 (\Rightarrow) to f|_{[c,b]} at the point d shows that the restriction f|_{[c,d]} belongs to \mathcal{R}[c,d] (if d = b, we already have f|_{[c,b]} \in \mathcal{R}[c,b]). ∎Two applications of the (\Rightarrow) direction of the Additivity Theorem 7.2.9 (T17). Proof of 7.2.11 (by induction on m; omitted in Bartle): For m = 1, [c_0,c_1] = [a,b], which is trivial; for m = 2, this is Theorem 7.2.9. Assume the result holds for m - 1 subintervals. Given a = c_0 < c_1 < \cdots < c_{m-1} < c_m = b, apply Theorem 7.2.9 at c = c_{m-1} \in (a,b): f is integrable on [a,c_{m-1}] and on [c_{m-1},c_m], with
\int_a^b f = \int_a^{c_{m-1}} f + \int_{c_{m-1}}^{c_m} f.
Applying the induction hypothesis to f|_{[a,c_{m-1}]} on a = c_0 < \cdots < c_{m-1} gives \int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f. ∎Mathematical induction on m using Theorem 7.2.9.   Important Exam Result · Integrable on every [c,b] (Bartle Ex 7.2.11–7.2.13; PYQ MAT-22 MCQ 24, 25, Q28) Statement (Bartle Ex 7.2.11): If f is bounded by M on [a,b] and f|_{[c,b]} \in \mathcal{R}[c,b] for every c \in (a,b), then f \in \mathcal{R}[a,b] and \lim_{c\to a^+} \int_c^b f = \int_a^b f. Proof: Given \varepsilon > 0, choose c \in (a,b) with c - a < \varepsilon/(4M). Define \alpha_c(x) := -M and \omega_c(x) := M on [a,c), and \alpha_c(x) := \omega_c(x) := f(x) on [c,b]. By Theorem 7.2.9 (\Leftarrow), \alpha_c, \omega_c \in \mathcal{R}[a,b], \alpha_c \le f \le \omega_c, and \int_a^b(\omega_c - \alpha_c) = \int_a^c 2M + \int_c^b 0 = 2M(c-a) < \varepsilon/2 < \varepsilon. By the Squeeze Theorem 7.2.3, f \in \mathcal{R}[a,b]. Moreover, \big\lvert\int_a^b f - \int_c^b f\big\rvert = \big\lvert\int_a^c f\big\rvert \le M(c-a) \to 0 as c \to a^+. Application to \sin(1/x) (Bartle Ex 7.2.12; PYQ MAT-22 MCQ 25): Let g(x) := \sin(1/x) for x \in (0,1] and g(0) := 0. Then \lvert g(x)\rvert \le 1 on [0,1] (bounded!), and for every c \in (0,1), g is continuous on [c,1] and hence in \mathcal{R}[c,1] by Theorem 7.2.7. By Exercise 7.2.11, g \in \mathcal{R}[0,1]. Why boundedness cannot be dropped (Bartle Ex 7.2.13): f(x) = 1/x on (0,1], f(0)=0 C16 is continuous (hence in \mathcal{R}[c,1]) for every c \in (0,1), but f is unbounded on [0,1], so f \notin \mathcal{R}[0,1]. In the model paper: Q27 (PYQ MAT-22 Q37b), Q28 (PYQ MAT-22 MCQ 24, 25, MAT-21 MCQ 18).
</section>
<!-- T19 (K19) --> <section class="page"> T19<h2>Oriented integrals and additivity for any three points</h2> Defining \int_\beta^\alpha f := -\int_\alpha^\beta f and \int_\alpha^\alpha f := 0 so additivity holds in any order (Bartle Definition 7.2.12, Theorem 7.2.13). Definition 7.2.12 & Theorem 7.2.13 (Bartle pp. 214–215, eq. (8)) · say them word for word Definition 7.2.12: If f \in \mathcal{R}[a,b] and if \alpha, \beta \in [a,b] with \alpha < \beta, we define

\int_\beta^\alpha f := -\int_\alpha^\beta f \qquad \text{and} \qquad \int_\alpha^\alpha f := 0.
 Theorem 7.2.13: If f \in \mathcal{R}[a,b] and if \alpha, \beta, \gamma are any numbers in [a,b], then
\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f,
in the sense that the existence of any two of these integrals implies the existence of the third and the equality. In plain words: integrating backwards from \beta to \alpha flips the sign, integrating from \alpha to \alpha gives 0, and with those two conventions \int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f holds no matter which of \alpha, \beta, \gamma is smallest, middle, or largest. Proof of Theorem 7.2.13 (Bartle pp. 214–215) If any two of the numbers \alpha, \beta, \gamma are equal, the formula \int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f holds immediately by Definition 7.2.12 (for instance, if \gamma = \beta, it reads \int_\alpha^\beta f = \int_\alpha^\beta f + 0; if \alpha = \beta, it reads 0 = \int_\alpha^\gamma f - \int_\alpha^\gamma f).Definition 7.2.12 handles all cases with repeated points. Assume \alpha, \beta, \gamma are all distinct. For any triple, define the cyclic sum L(\alpha, \beta, \gamma) := \int_\alpha^\beta f + \int_\beta^\gamma f + \int_\gamma^\alpha f. Since \int_\gamma^\alpha f = -\int_\alpha^\gamma f, the desired equation \int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f holds if and only if L(\alpha, \beta, \gamma) = 0.Move \int_\alpha^\gamma f + \int_\gamma^\beta f to the left-hand side using \int_\alpha^\gamma f = -\int_\gamma^\alpha f. When \alpha < \gamma < \beta, the Additivity Theorem 7.2.9 gives \int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f, so L(\alpha, \beta, \gamma) = 0.Theorem 7.2.9 (T17) for the standard ordering \alpha < \gamma < \beta. Cycling the three letters leaves the sum unchanged: L(\beta, \gamma, \alpha) = L(\gamma, \alpha, \beta) = L(\alpha, \beta, \gamma). Swapping two adjacent letters reverses the orientation of every integral and negates the sum: L(\beta, \alpha, \gamma) = L(\alpha, \gamma, \beta) = L(\gamma, \beta, \alpha) = -L(\alpha, \beta, \gamma).Commutativity of addition and \int_u^v f = -\int_v^u f. Every one of the 6 possible orderings of three distinct numbers is obtained from \alpha < \gamma < \beta by a cyclic shift or a swap, so L(\alpha, \beta, \gamma) = 0 for all configurations of \alpha, \beta, \gamma. ∎0 and -0 are both 0.   Worked example · Numerical check for \alpha = 3, \beta = 1, \gamma = 0 (Model Paper Q9) Let f(x) = x on [0,3], and take \alpha = 3, \beta = 1, \gamma = 0 (so \gamma < \beta < \alpha). Using \int_a^b x\,dx = \tfrac12(b^2 - a^2) and Definition 7.2.12: <ul> Left-hand side: \displaystyle\int_3^1 x\,dx = -\int_1^3 x\,dx = -\tfrac12(3^2 - 1^2) = -4. Right-hand side: \displaystyle\int_3^0 x\,dx + \int_0^1 x\,dx = -\int_0^3 x\,dx + \int_0^1 x\,dx = -\tfrac92 + \tfrac12 = -4. Both sides equal -4. In the model paper: Q9. </section>
=== NOTES / DOUBTS (STAGE 2 OF 6) ===

- Completed Stage 2 (src/m3/prepare_03_sec72.html): covers Unit 2 (Bartle §7.2, topics T10–T19 corresponding to K10–K19), including all figure placeholders (F08_dirichlet_tags, F05_squeeze_step, F16_elementary_steps, F06_cont_min_max_steps, F07_monotone_staircase, F13_additivity_split), word-for-word theorems, hypotheses boxes with counterexamples, skeletons, scratch work, step-by-step proofs with .why justifications, and the known issue on Theorem 7.2.9 (M>0 bound from 7.1.6).

- Next up in Stage 3: src/m3/prepare_04_sec73.html (Unit 3: Bartle §7.3, topics T20–T32, plus Counterexample library page, "Which tool?" mixed drill, Theorem sheet, Recall plan, and PYQ → topic table).
- Reminder for Stage 6: Please paste or attach figs.py and figs_m2.py before or when requesting Stage 6 so src/m3/figs_m3.py can be generated using the exact S and Plot classes.